SMART INDIA HACKATHON 2026

TARAL

Thermal Analysis and Reconstruction using AI for Ocean Layers

PS ID:	26066
PS:	OceanEmbed – Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations.
Category:	Software
Theme:	Disaster Management

TEAM B2S – BHARAT THROUGH SPACE

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Chapter 1

Project Information

1.1 Problem Statement

The problem statement, identified as Problem Statement ID 26066 under the Smart India Hackathon (SIH) 2026, is titled:

> *OceanEmbed – Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations*

The objective is to develop a satellite embedding-based deep learning framework for reconstructing depth-wise subsurface ocean temperature from daily surface observations at a spatial resolution of 0.25°. The study focuses on the North Indian Ocean, covering 5°N to 30°N latitude and 45°E to 105°E longitude.

Subsurface temperature is important for understanding ocean circulation, heat content, stratification, climate variability, air-sea interaction, and marine ecosystems. However, direct measurements are spatially and temporally limited and primarily rely on in-situ systems such as ARGO floats, moored buoys, gliders, and ships.

Satellite observations provide continuous, large-scale information about surface conditions. Variables such as Sea Surface Temperature (SST), Sea Surface Salinity (SSS), Sea Level Anomaly (SLA), surface currents, and winds provide indirect information about subsurface processes.

The proposed framework aims to learn the relationship between these observable surface conditions and subsurface temperature using satellite embeddings and deep learning. The temperature is reconstructed at standard depth levels from the surface to 1000 m.

1.2 Background

Subsurface ocean temperature is a fundamental variable for understanding ocean circulation, upper-ocean heat content, stratification, climate variability, air-sea interaction, and

marine ecosystems. Accurate vertical temperature profiles are important for applications such as marine heatwave monitoring, fisheries, and data assimilation.

However, direct subsurface measurements are spatially and temporally limited. They mainly rely on in-situ systems such as ARGO profiling floats, moored buoys, gliders, and ship-based observations, which cannot provide continuous basin-scale temperature fields.

Satellite observations offer continuous, large-scale information about surface ocean conditions. Variables such as Sea Surface Temperature (SST), Sea Surface Salinity (SSS), Sea Level Anomaly (SLA), surface currents, and winds contain indirect signatures of subsurface processes through mechanisms such as thermocline displacement, mesoscale eddies, vertical mixing, transport, and ocean-atmosphere coupling.

Recent advances in Artificial Intelligence (AI), Deep Learning (DL), and representation learning enable these complex relationships to be learned from data. Satellite embeddings can transform multidimensional surface observations into compact latent representations that capture relevant ocean dynamics and nonlinear relationships between surface conditions and subsurface temperature structure.

1.3 Why we chose this problem

The OceanEmbed problem was selected because subsurface ocean temperature is important for understanding ocean dynamics, climate variability, air-sea interactions, and marine ecosystems, while continuous large-scale subsurface observations remain limited.

The problem offers an opportunity to address this limitation using surface satellite observations such as Sea Surface Temperature (SST), Sea Surface Salinity (SSS), Sea Level Anomaly (SLA), surface currents, and winds, which contain information about the physical state and dynamics of the ocean.

The availability of multiple complementary ocean datasets also enables the development of a unified data-driven framework using Artificial Intelligence and Deep Learning to learn relationships between surface conditions and subsurface temperature.

The problem is also practically relevant, with potential applications in marine heatwave monitoring, fisheries, data assimilation, ocean circulation, and climate studies. Thus, OceanEmbed combines an important oceanographic challenge with modern deep learning techniques to develop a useful reconstruction framework.

1.4 Problem Definition

The OceanEmbed project aims to reconstruct the vertical structure of subsurface ocean temperature using daily surface ocean observations. The inputs include Sea Surface Temperature (SST), Sea Surface Salinity (SSS), Sea Level Anomaly (SLA), surface current components, and surface wind components.

The study covers the North Indian Ocean, from 5°N to 30°N and 45°E to 105°E, at a spatial resolution of 0.25° with daily observations.

The objective is to learn the relationship between surface observations and subsurface temperature using a satellite embedding-based deep learning framework. The model reconstructs temperature at 15 standard depths from the surface to 1000 m:

$$0,\ 5,\ 10,\ 20,\ 30,\ 50,\ 75,\ 100,\ 125,\ 150,\ 200,\ 300,\ 500,\ 700,\ 1000 \text{ m}$$

Thus, the problem is formulated as a supervised learning task in which multidimensional surface ocean observations are used to estimate the corresponding depth-wise subsurface temperature field.

1.5 Study Region

The OceanEmbed framework focuses on the North Indian Ocean region, defined by the geographical boundaries specified in the problem statement:

- **Latitude:** 5°N to 30°N
- **Longitude:** 45°E to 105°E

The region includes the Arabian Sea, Bay of Bengal, and surrounding North Indian Ocean waters.

The reconstruction uses a spatial resolution of $0.25° \times 0.25°$ with daily surface observations as model inputs.

1.6 Objectives

The main objective of OceanEmbed is to develop a deep learning-based framework for reconstructing subsurface ocean temperature from surface satellite observations.

The specific objectives are:

1. To collect and integrate relevant surface and subsurface ocean observation datasets.
2. To preprocess and harmonize the datasets across spatial, temporal, and variable dimensions.
3. To develop a satellite embedding-based approach for extracting meaningful representations from multidimensional surface observations.
4. To develop a deep learning framework that learns the relationship between surface observations and subsurface temperature structure.

5. To reconstruct subsurface temperature at 15 standard depths from the surface to 1000 m.

6. To generate daily reconstructed temperature fields at $0.25^\circ \times 0.25^\circ$ resolution for the North Indian Ocean.

7. To evaluate the reconstructed fields using suitable metrics and independent ARGO observations.

8. To demonstrate the potential of the framework for marine heatwave monitoring, fisheries, data assimilation, ocean circulation, and climate studies. ‘‘‘

1.7 Applications

The reconstructed subsurface ocean temperature fields generated by OceanEmbed can support several oceanographic and environmental applications:

- **Marine Heatwave Monitoring:** Understanding the vertical extent and evolution of marine heatwaves.
- **Fisheries:** Assessing ocean conditions that influence marine ecosystems and fisheries.
- **Data Assimilation:** Providing additional temperature information for ocean data assimilation systems.
- **Ocean Circulation Studies:** Studying temperature distribution, circulation, and related physical processes.
- **Climate Variability:** Analyzing subsurface ocean changes associated with climate variability.
- **Air–Sea Interaction:** Supporting the study of interactions between the ocean and atmosphere.
- **Marine Ecosystems:** Understanding environmental conditions affecting marine ecosystems.

1.8 Expected Outcomes

The expected outcomes of the OceanEmbed project are:

1. A harmonized dataset containing the required surface and subsurface ocean observations for the North Indian Ocean.

2. A standardized preprocessing pipeline to handle differences in spatial and temporal resolution, formats, and dataset characteristics.

3. A satellite embedding component for generating meaningful representations from multidimensional surface observations.

4. A deep learning framework for reconstructing subsurface ocean temperature from surface observations.

5. Daily subsurface temperature fields at $0.25^\circ \times 0.25^\circ$ spatial resolution.

6. Reconstructed temperature profiles at 15 standard depths from the surface to 1000 m.

7. Validation of the reconstructed fields using suitable evaluation metrics and independent ARGO observations.

8. A proof-of-concept implementation demonstrating the feasibility of subsurface temperature reconstruction from surface observations.

Chapter 2

Proposed Solution

2.1 Overview of the Proposed Solution

The proposed OceanEmbed framework aims to reconstruct subsurface ocean temperature from surface ocean observations using a satellite embedding-based deep learning approach.

Multiple ocean observation datasets are collected, processed, and harmonized to ensure compatibility across their spatial, temporal, and variable characteristics. Surface variables, including Sea Surface Temperature (SST), Sea Surface Salinity (SSS), Sea Level Anomaly (SLA), surface currents, and surface winds, are used as model inputs.

These multidimensional observations are transformed into meaningful representations through a satellite embedding component. The resulting embeddings are then provided to a deep learning reconstruction model that learns the relationship between surface ocean conditions and subsurface temperature structure.

The framework produces depth-wise subsurface temperature fields for the North Indian Ocean at $0.25^\circ \times 0.25^\circ$ spatial resolution, covering 15 standard depths from the surface to 1000 m. The reconstructed fields are evaluated using suitable metrics and independent ARGO observations.

Overall, OceanEmbed provides a data-driven framework for estimating subsurface ocean temperature from readily available surface observations.

2.2 Data Collection and Integration

The OceanEmbed framework requires multiple ocean observation datasets to provide complementary information about the ocean surface and subsurface. The selected datasets include DUACS, GlobCurrent, ARGO, CCMP, GLORYS, OSCAR, OISST, and OISSS.

These datasets differ in oceanographic variables, spatial and temporal resolutions, coverage periods, and file formats. Therefore, the required datasets will first be collected

and their characteristics examined before integration.

The datasets will provide variables such as Sea Surface Temperature (SST), Sea Surface Salinity (SSS), Sea Surface Height or Sea Level Anomaly, surface currents, surface winds, and subsurface temperature observations. ARGO and other subsurface datasets will provide temperature information required for training and validation.

Since the datasets may use different coordinate systems, time representations, spatial grids, and formats, preprocessing and harmonization are required before integration. The datasets will therefore be aligned to a common representation suitable for the deep learning framework.

The integration process will produce a consistent collection of surface and subsurface observations for the selected North Indian Ocean region.

2.3 Data Preprocessing and Harmonization

The collected datasets come from different sources and may differ in spatial grids, temporal resolutions, coordinate conventions, units, file formats, and data coverage. Therefore, preprocessing and harmonization are required before integration to ensure consistent model inputs.

The main preprocessing steps include:

1. **Data Inspection:** Examine the structure, variables, dimensions, coordinates, temporal coverage, and metadata of each dataset.

2. **Quality Checking:** Identify missing values, invalid values, duplicates, and other data-quality issues.

3. **Spatial Harmonization:** Transform datasets to a common spatial representation for the selected North Indian Ocean region at 0.25° resolution.

4. **Temporal Harmonization:** Align observations to a common temporal representation based on daily observations.

5. **Variable Standardization:** Ensure consistent units, naming conventions, and variable representations.

6. **Handling Missing Data:** Identify and appropriately handle missing observations while preserving data reliability.

7. **Data Alignment:** Match spatial and temporal observations across datasets for use during model development.

The final preprocessing procedure will be determined through detailed dataset auditing. Dataset-specific transformations and decisions will be documented to maintain reproducibility.

2.4 Satellite Embedding Engine

The satellite embedding engine is a key component of the OceanEmbed framework. It transforms multidimensional surface ocean observations into meaningful numerical representations called embeddings.

The input consists of harmonized surface variables such as Sea Surface Temperature (SST), Sea Surface Salinity (SSS), Sea Surface Height (SSH) or Sea Level Anomaly (SLA), surface currents, and surface winds.

Instead of directly providing all raw variables to the reconstruction model, the embedding stage learns compact representations that capture important spatial and physical patterns in the surface observations. These representations help the deep learning model learn relationships between surface ocean conditions and subsurface temperature structure.

The embedding process is designed to preserve relevant information while providing a suitable representation for the reconstruction model. The exact embedding architecture and feature extraction approach will be selected during model development based on experimentation and evaluation.

The generated embeddings are then provided to the deep learning reconstruction component to estimate depth-wise subsurface ocean temperature.

2.5 Deep Learning Reconstruction

The deep learning reconstruction component estimates subsurface ocean temperature from the representations generated from surface ocean observations.

The harmonized surface variables and their embeddings are provided as inputs to learn the relationship between surface ocean conditions and the vertical temperature structure.

The reconstruction is performed at 15 predefined depth levels from the surface to 1000 m:

$$0,\ 5,\ 10,\ 20,\ 30,\ 50,\ 75,\ 100,\ 125,\ 150,\ 200,\ 300,\ 500,\ 700,\ 1000 \text{ m}$$

The model will capture spatial patterns and nonlinear relationships within the ocean observations. Candidate approaches include Convolutional Neural Networks (CNNs), Vision Transformers, Autoencoders, Graph Neural Networks (GNNs), and attention-based hybrid architectures.

The final architecture will be selected during model development based on suitability for the reconstruction task and validation performance.

The output will be a depth-wise subsurface ocean temperature field for the selected North Indian Ocean region.

2.6 Validation and Evaluation

The reconstructed subsurface ocean temperature fields will be evaluated to assess the accuracy and reliability of the OceanEmbed framework.

Independent subsurface observations, particularly ARGO observations, will be used for validation. Reconstructed temperatures will be compared with observed values at matching spatial locations, dates, and depth levels wherever suitable observations are available.

The evaluation will use statistical measures such as Root Mean Square Error (RMSE), correlation, and bias to quantify agreement between reconstructed and observed temperatures.

The validation process will assess the accuracy of the reconstructed vertical temperature structure and identify variations in performance across different depth levels and regions.

The final evaluation procedure and specific metrics will be determined during the model development and validation stage.

2.7 Overall Workflow

The OceanEmbed framework consists of sequential stages that transform raw ocean observation datasets into reconstructed subsurface temperature fields.

The overall workflow includes:

1. **Dataset Collection:** Collect the selected ocean observation datasets required for the project.

2. **Dataset Audit and Quality Checking:** Examine variables, spatial and temporal coverage, formats, missing values, and other relevant characteristics.

3. **Preprocessing and Harmonization:** Standardize spatial grids, temporal representation, units, coordinate conventions, and other required characteristics.

4. **Data Integration:** Align surface and subsurface observations to create a consistent dataset for model development.

5. **Satellite Embedding:** Transform multidimensional surface observations into meaningful numerical representations.

6. **Deep Learning Reconstruction:** Learn the relationship between surface observations and subsurface ocean temperature using the generated representations.

7. **Temperature Reconstruction:** Generate depth-wise subsurface temperature estimates from the surface to 1000 m.

8. **Validation and Evaluation:** Compare reconstructed temperature fields with independent observations such as ARGO data using suitable statistical metrics.

The overall workflow can be summarized as:

Dataset Collection → Audit and Quality Checking → Preprocessing and Harmonization → Data Integration → Satellite Embedding → Deep Learning Reconstruction → Temperature Reconstruction → Validation and Evaluation

Chapter 3

Datasets

3.1 OISST Dataset

3.1.1 Dataset Identity

Dataset Name	NOAA 1/4° Daily Optimum Interpolation Sea Surface Temperature
Short Name / Acronym	OISST
Version / Edition	Version 2.1
Provider / Organization	NOAA / National Centers for Environmental Information (NCEI)
Official Source / Landing Page	
Dataset DOI / Citation	10.25921/RE9P-PT57

3.1.2 Temporal Coverage

Actual Start Date	1981-09-01
Actual End Date / Current Date	Present
Our Selected Time Range	
Temporal Resolution	Daily
Temporal Sampling / Cadence	1 day

3.1.3 Spatial Characteristics

Spatial Coverage	Global
Actual Spatial Grid	$0.25° \times 0.25°$
Latitude Range	
Longitude Range / Convention	
Depth / Vertical Coverage	Sea surface

3.1.4 Variables

Variables Collected	sst, anom, err, ice
Variable Actually Used	
Units	SST, anomaly and error: °C; ice: fraction

3.1.5 Acquisition

Collected Through	NOAA / NCEI
Collected Using	
Acquisition Script	
Acquisition Date	
Collection Time	
Retry / Failure Notes	

3.1.6 Storage

Estimated Raw Data Size	
Number of Raw Files	
Raw File Format	NetCDF / NetCDF-4

3.1.7 Processing

Processing / Standardization Steps	
Project Output Format	
Size After Processing	
Processing Time	
Final Shape / Dimensions	

3.1.8 Quality Control

Quality-Control / Flags	Quality-controlled optimum-interpolation product with bias correction
Missing Data Behavior	
Known Limitations / Biases	

3.1.9 Integration in OceanEmbed

Main Purpose in OceanEmbed	Sea surface temperature input
Model Stage / Component	Surface input / embedding stage
Fusion With	SSS, SSH/SLA, currents and winds
Alignment Method	

3.1.10 Dataset Selection Decisions

Why This Dataset Was Chosen	Provides long-term daily SST at the required 0.25° resolution
Important Decisions Made	
Alternatives Considered	

3.1.11 Audit Findings

Important Findings During Audit	
Important Findings During Collection	

3.1.12 Reproducibility

Exact Download Pattern / Endpoint	
Checksums / Validation	
Repository Path	

3.1.13 Evidence

Provider Documentation URL	
Project Audit Evidence	

3.2 GlobCurrent Dataset

3.2.1 Dataset Identity

Field	Details
Dataset Name	Global Total (COPERNICUS-GLOBCURRENT), Ekman and Geostrophic Currents at Surface and 15m
Short Name / Acronym	GlobCurrent
Version / Edition	
Provider / Organization	Copernicus Marine Service
Official Source / Landing Page	
Product ID	MULTIOBS_GLO_PHY_MYNRT_015_003
DOI	10.48670/mds-00327
Official Source	Copernicus Marine Service

3.2.2 Temporal Coverage

Field	Details
Actual Start Date	1993-01-01
Actual End Date / Current Date	Present
Our Selected Time Range	
Temporal Resolution	Hourly / Daily / Monthly
Temporal Sampling / Cadence	

3.2.3 Spatial Characteristics

Field	Details
Spatial Coverage	Global
Actual Spatial Grid	$0.25^\circ \times 0.25^\circ$
Latitude Range	Approximately -89.87° to 89.88°
Longitude Range / Convention	Approximately -179.87° to 179.88°
Depth / Vertical Coverage	Surface and 15 m

3.2.4 Variables

Field	Details
Variables Collected	Eastward and Northward Sea Water Velocity; Ekman and Geostrophic Current Components
Variable Actually Used	
Units	m/s

3.2.5 Acquisition

Field	Details
Collected Through	Copernicus Marine Service
Collected Using	
Acquisition Script	
Acquisition Date	
Collection Time	
Retry / Failure Notes	

3.2.6 Storage

Field	Details
Estimated Raw Data Size	
Number of Raw Files	
Raw File Format	NetCDF-4

3.2.7 Processing

Field	Details
Processing / Standardization Steps	
Project Output Format	
Size After Processing	
Processing Time	
Final Shape / Dimensions	

3.2.8 Quality Control

Field	Details
Quality-Control / Flags	
Missing Data Behavior	
Known Limitations / Biases	

3.2.9 Integration in OceanEmbed

Field	Details
Main Purpose in OceanEmbed	Surface ocean-current input
Model Stage / Component	
Fusion With	
Alignment Method	

3.2.10 Selection Decisions

Field	Details
Why This Dataset Was Chosen	
Important Decisions Made	
Alternatives Considered	

3.2.11 Audit Findings

Field	Details
Important Findings During Audit	
Important Findings During Collection	

3.2.12 Reproducibility

Field	Details
Exact Download Pattern / Endpoint	
Checksums / Validation	
Repository Path	

3.2.13 Evidence

Field	Details
Provider Documentation URL	
Project Audit Evidence	

3.3 ARGO Dataset

3.3.1 Dataset Identity

Dataset Name	Argo Float Data and Metadata
Short Name / Acronym	ARGO
Version / Edition	
Provider / Organization	International Argo Program / Argo Global Data Assembly Centres (GDAC)
Official Source / Landing Page	https://argo.ucsd.edu/data/data-access/
Dataset DOI / Citation	10.17882/42182

3.3.2 Temporal Coverage

Actual Start Date	Approximately 2000
Actual End Date / Current Date	Present
Our Selected Time Range	
Temporal Resolution	Profile / float-cycle dependent
Temporal Sampling / Cadence	Typically approximately 10 days per float cycle

3.3.3 Spatial Characteristics

Spatial Coverage	Global ocean
Actual Spatial Grid	Irregular point profiles
Latitude Range	
Longitude Range / Convention	
Depth / Vertical Coverage	Surface to deep ocean profiles

3.3.4 Variables

Variables Collected	Pressure, Temperature, Salinity, QC Flags, Adjusted Temperature, Adjusted Salinity
Variable Actually Used	
Units	Pressure: dbar; Temperature: °C; Salinity: PSU / practical salinity scale

3.3.5 Acquisition

Collected Through	Argo Global Data Assembly Centres (GDAC)
Collected Using	
Acquisition Script	
Acquisition Date	
Collection Time	
Retry / Failure Notes	

3.3.6 Storage

Estimated Raw Data Size	
Number of Raw Files	
Raw File Format	NetCDF

3.3.7 Processing

Processing / Standardization Steps	
Project Output Format	
Size After Processing	
Processing Time	
Final Shape / Dimensions	

3.3.8 Quality Control

Quality-Control / Flags	Argo quality-control flags are provided with profile measurements
Missing Data Behavior	
Known Limitations / Biases	Irregular spatial and temporal sampling

3.3.9 Integration in OceanEmbed

Main Purpose in OceanEmbed	Independent reference and validation of reconstructed subsurface temperature
Model Stage / Component	Validation
Fusion With	Surface observations and reconstructed temperature
Alignment Method	

3.3.10 Dataset Selection Decisions

Why This Dataset Was Chosen	Provides in-situ subsurface ocean temperature profiles for independent validation
Important Decisions Made	
Alternatives Considered	

3.3.11 Audit Findings

Important Findings During Audit	
Important Findings During Collection	

3.3.12 Reproducibility

Exact Download Pattern / Endpoint	
Checksums / Validation	
Repository Path	

3.3.13 Evidence

Provider Documentation URL	
Project Audit Evidence	

3.4 CCMP Dataset

3.4.1 Dataset Identity

Dataset Name	RSS CCMP 6-Hourly 10 Meter Surface Winds Level 4 Version 3.1
Short Name / Acronym	CCMP
Version / Edition	Version 3.1
Provider / Organization	Remote Sensing Systems (RSS)
Official Source / Landing Page	
Dataset DOI / Citation	10.5067/CCMP-6HW10M-L4V31
Product ID	CCMP_WINDS_10M6HR_L4_V3.1

Temporal Coverage

Actual Start Date	1993-01-01
Actual End Date / Current Date	Present
Our Selected Time Range	
Temporal Resolution	6-hourly
Temporal Sampling / Cadence	4 observations per day; one NetCDF-4 file per day

3.4.2 Spatial Characteristics

Spatial Coverage	Near-global
Actual Spatial Grid	$0.25° \times 0.25°$
Latitude Range	80°S to 80°N
Longitude Range / Convention	$-180°$ to $180°$
Depth / Vertical Coverage	10 m surface wind

3.4.3 Variables

Variables Collected	uwnd, vwnd, ws, nobs, latitude, longitude, time
Variable Actually Used	
Units	Wind components and wind speed: m/s

3.4.4 Acquisition

Collected Through	Remote Sensing Systems
Collected Using	
Acquisition Script	
Acquisition Date	
Collection Time	
Retry / Failure Notes	

3.4.5 Storage

Estimated Raw Data Size	
Number of Raw Files	
Raw File Format	NetCDF-4

3.4.6 Processing

Processing / Standardization Steps	
Project Output Format	
Size After Processing	
Processing Time	
Final Shape / Dimensions	

3.4.7 Quality Control

Quality-Control / Flags	Observation count information is provided through nobs
Missing Data Behavior	
Known Limitations / Biases	Near-global spatial coverage does not include regions beyond approximately 80° latitude

3.4.8 Integration in OceanEmbed

Main Purpose in OceanEmbed	Surface wind input
Model Stage / Component	Surface input / embedding stage
Fusion With	SST, SSS, SSH/SLA and surface currents
Alignment Method	

3.4.9 Dataset Selection Decisions

Why This Dataset Was Chosen	Provides 10 m surface wind components at 0.25° resolution
Important Decisions Made	CCMP is used as the project wind dataset instead of ERA5
Alternatives Considered	ERA5

3.4.10 Audit Findings

Important Findings During Audit	
Important Findings During Collection	

3.4.11 Reproducibility

Exact Download Pattern / Endpoint	
Checksums / Validation	
Repository Path	

3.4.12 Evidence

Provider Documentation URL	
Project Audit Evidence	

3.5 GLORYS Dataset

3.5.1 Dataset Identity

Dataset Name	CMEMS Global Ocean Ensemble Physics Reanalysis
Short Name / Acronym	GLORYS
Version / Edition	
Provider / Organization	Copernicus Marine Service
Official Source / Landing Page	
Dataset DOI / Citation	
Product ID	GLOBAL_MULTIYEAR_PHY_ENS_001_031

3.5.2 Temporal Coverage

Actual Start Date	1993
Actual End Date / Current Date	2024
Our Selected Time Range	
Temporal Resolution	Daily / Monthly
Temporal Sampling / Cadence	

3.5.3 Spatial Characteristics

Spatial Coverage	Global ocean
Actual Spatial Grid	$0.25^\circ \times 0.25^\circ$
Latitude Range	
Longitude Range / Convention	
Depth / Vertical Coverage	75 vertical levels

3.5.4 Variables

Variables Collected	Potential temperature (thetao), salinity (so), eastward velocity (uo), northward velocity (vo), sea surface height and other ocean-physics variables
Variable Actually Used	
Units	Temperature: °C; salinity: PSU; velocity: m/s; SSH: m

3.5.5 Acquisition

Collected Through	Copernicus Marine Service
Collected Using	
Acquisition Script	
Acquisition Date	
Collection Time	
Retry / Failure Notes	

3.5.6 Storage

Estimated Raw Data Size	
Number of Raw Files	
Raw File Format	NetCDF-4

3.5.7 Processing

Processing / Standardization Steps	
Project Output Format	
Size After Processing	
Processing Time	
Final Shape / Dimensions	

3.5.8 Quality Control

Quality-Control / Flags	
Missing Data Behavior	
Known Limitations / Biases	

3.5.9 Integration in OceanEmbed

Main Purpose in OceanEmbed	Ocean-state and subsurface reference information
Model Stage / Component	
Fusion With	
Alignment Method	

3.5.10 Dataset Selection Decisions

Why This Dataset Was Chosen	
Important Decisions Made	
Alternatives Considered	

3.5.11 Audit Findings

Important Findings During Audit	
Important Findings During Collection	

3.5.12 Reproducibility

Exact Download Pattern / Endpoint	
Checksums / Validation	
Repository Path	

3.5.13 Evidence

Provider Documentation URL	
Project Audit Evidence	

3.6 OSCAR Dataset

3.6.1 Dataset Identity

Dataset Name	Ocean Surface Current Analyses Real-time (OSCAR) Surface Currents – Final 0.25 Degree
Short Name / Acronym	OSCAR
Version / Edition	Version 2.0
Provider / Organization	NASA PO.DAAC / Earth & Space Research (ESR)
Official Source / Landing Page	
Dataset DOI / Citation	10.5067/OSCAR-25F20
Product ID	OSCAR_L4_OC_FINAL_V2.0

3.6.2 Temporal Coverage

Actual Start Date	1993-01-01
Actual End Date / Current Date	2026-01-17
Our Selected Time Range	
Temporal Resolution	Daily
Temporal Sampling / Cadence	1 day

3.6.3 Spatial Characteristics

Spatial Coverage	Global
Actual Spatial Grid	$0.25^\circ \times 0.25^\circ$
Latitude Range	-89.75° to 89.75°
Longitude Range / Convention	-180° to 180°
Depth / Vertical Coverage	Surface current representative of the upper approximately 30 m mixed layer

3.6.4 Variables

Variables Collected	u, v, ug, vg
Variable Actually Used	
Units	m/s

3.6.5 Acquisition

Collected Through	NASA PO.DAAC
Collected Using	
Acquisition Script	
Acquisition Date	
Collection Time	
Retry / Failure Notes	

3.6.6 Storage

Estimated Raw Data Size	
Number of Raw Files	
Raw File Format	NetCDF-4

3.6.7 Processing

Processing / Standardization Steps	
Project Output Format	
Size After Processing	
Processing Time	
Final Shape / Dimensions	

3.6.8 Quality Control

Quality-Control / Flags	Final, interim and near-real-time product levels are available
Missing Data Behavior	
Known Limitations / Biases	Surface current estimates are derived using physical relationships involving sea surface height, winds and SST gradients

3.6.9 Integration in OceanEmbed

Main Purpose in OceanEmbed	Surface ocean-current input
Model Stage / Component	Surface input / embedding stage
Fusion With	Other surface ocean and atmospheric variables
Alignment Method	

3.6.10 Dataset Selection Decisions

Why This Dataset Was Chosen	Provides global daily surface current information at 0.25° resolution
Important Decisions Made	
Alternatives Considered	GlobCurrent

3.6.11 Audit Findings

Important Findings During Audit	
Important Findings During Collection	

3.6.12 Reproducibility

Exact Download Pattern / Endpoint	
Checksums / Validation	
Repository Path	

3.6.13 Evidence

Provider Documentation URL	
Project Audit Evidence	

3.7 OISSS Dataset

3.7.1 Dataset Identity

Dataset Name	Multi-Mission Optimally Interpolated Sea Surface Salinity Global Dataset V2
Short Name / Acronym	OISSS
Version / Edition	Version 2
Provider / Organization	NASA / Jet Propulsion Laboratory (JPL) / PO.DAAC
Official Source / Landing Page	
Dataset DOI / Citation	
Product ID	OISSS_L4_multimission_7day_v2

3.7.2 Temporal Coverage

Actual Start Date	2011-08-24
Actual End Date / Current Date	Present
Our Selected Time Range	
Temporal Resolution	4-day
Temporal Sampling / Cadence	

3.7.3 Spatial Characteristics

Spatial Coverage	Global
Actual Spatial Grid	$0.25° \times 0.25°$
Latitude Range	
Longitude Range / Convention	
Depth / Vertical Coverage	Sea surface

3.7.4 Variables

Variables Collected	sss, sss_empirical_uncertainty, sss_formal_uncertainty
Variable Actually Used	
Units	10^{-3}

3.7.5 Acquisition

Collected Through	NASA PO.DAAC / Earthdata
Collected Using	
Acquisition Script	
Acquisition Date	
Collection Time	
Retry / Failure Notes	

3.7.6 Storage

Estimated Raw Data Size	
Number of Raw Files	
Raw File Format	NetCDF-4

3.7.7 Processing

Processing / Standardization Steps	
Project Output Format	
Size After Processing	
Processing Time	
Final Shape / Dimensions	

3.7.8 Quality Control

Quality-Control / Flags	Empirical and formal uncertainty variables are provided
Missing Data Behavior	
Known Limitations / Biases	

3.7.9 Integration in OceanEmbed

Main Purpose in OceanEmbed	Sea surface salinity input
Model Stage / Component	Surface input / embedding stage
Fusion With	SST, SSH/SLA, currents and winds
Alignment Method	

3.7.10 Dataset Selection Decisions

Why This Dataset Was Chosen	Provides global multi-mission sea surface salinity at 0.25° resolution
Important Decisions Made	Version 2 is used as the project OISSS product
Alternatives Considered	

3.7.11 Audit Findings

Important Findings During Audit	
Important Findings During Collection	

3.7.12 Reproducibility

Exact Download Pattern / Endpoint	
Checksums / Validation	
Repository Path	

3.7.13 Evidence

Provider Documentation URL	
Project Audit Evidence	

3.7.14 Open Questions / Follow-ups

Open Questions / Follow-ups

3.8 DUACS Dataset

3.8.1 Dataset Identity

Field	Details
Dataset Name	Global Ocean Gridded L4 Sea Surface Heights and Derived Variables Reprocessed
Short Name / Acronym	DUACS
Version / Edition	DT-2024
Provider / Organization	Copernicus Marine Service / Mercator Ocean International
Official Source / Landing Page	https://data.marine.copernicus.eu/product/SEALEVEL_GLO_PHY_L4_MY_008_047/description
Dataset DOI / Citation	10.48670/moi-00148

3.8.2 Temporal Coverage

Actual Start Date	1993
Actual End Date / Current Date	Present
Our Selected Time Range	
Temporal Resolution	Daily
Temporal Sampling / Cadence	1 day

3.8.3 Spatial Characteristics

Spatial Coverage	Global
Actual Spatial Grid	$0.125^\circ \times 0.125^\circ$
Latitude Range	
Longitude Range / Convention	
Depth / Vertical Coverage	Surface variables

3.8.4 Variables

Variables Collected	Sea Surface Height, Sea Level Anomaly (SLA), Absolute Dynamic Topography (ADT), Geostrophic Eastward Velocity, Geostrophic Northward Velocity
Variable Actually Used	
Units	Sea level: m; velocity: m/s

3.8.5 Acquisition

Collected Through	Copernicus Marine Service
Collected Using	
Acquisition Script	
Acquisition Date	
Collection Time	
Retry / Failure Notes	

3.8.6 Storage

Estimated Raw Data Size	
Number of Raw Files	
Raw File Format	NetCDF-4

3.8.7 Processing

Processing / Standardization Steps	
Project Output Format	
Size After Processing	
Processing Time	
Final Shape / Dimensions	

3.8.8 Quality Control

Quality-Control / Flags	
Missing Data Behavior	
Known Limitations / Biases	

3.8.9 Integration in OceanEmbed

Main Purpose in OceanEmbed	Surface sea-level and current-related input
Model Stage / Component	
Fusion With	
Alignment Method	

3.8.10 Dataset Selection Decisions

Why This Dataset Was Chosen
Important Decisions Made
Alternatives Considered

3.8.11 Audit Findings

Important Findings During Audit
Important Findings During Collection

3.8.12 Reproducibility

Exact Download Pattern / Endpoint
Checksums / Validation
Repository Path

3.8.13 Evidence

Provider Documentation URL	https://data.marine.copernicus.eu/product/SEALEVEL_GLO_PHY_L4_MY_008_047/description
Project Audit Evidence	